

Ministry of Heavy Industries Summer Internship



सत्यमेव जयते

भारी उद्योग मंत्रालय MINISTRY OF **HEAVY INDUSTRIES**

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Declaration

I, **Dhruv Garg**, solemnly declare that this report titled “**IGNITE: Incentivizing Green New-age Infrastructure and Technology for EVs**” represents my original work and has not been submitted in whole or in part for any academic or professional qualification, except where duly acknowledged. All sources of information have been acknowledged, and verbatim extracts are indicated by quotation marks. Any views, opinions, or findings within this report are those of the author and do not necessarily reflect the views of the organization or institution associated with the internship.

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Index

Abstract

1.0 Objective

- 1.1 Evaluate FAME-III's Potential Impact
- 1.2 Analyze FAME-II Performance
- 1.3 Forecast Electric Two-Wheeler Sales
- 1.4 Identify Market Challenges and Opportunities
- 1.5 Propose Policy and Subsidy Recommendations
- 1.6 Consider Social and Environmental Implications
- 1.7 Recommend Consumer Awareness Initiatives

2.0 Methodology

- 2.1 Data Collection and Analysis
- 2.2 Market Analysis
- 2.3 Predictive Modeling
 - 2.3.1 Model One
 - 2.3.2 Model Two
 - 2.3.3 Model Three

3.0 Benefits and Recommendations for the Ministry

- 3.1 Accurate Sales Prediction (Model One)
- 3.2 Second-Hand EV Sales Prediction (Model Two)
- 3.3 Analysis of EV Vehicles (Model Three)
- 3.4 Overall Recommendations for the Ministry

4.0 Findings and Results

- 4.1 Electrification Target by 2030
- 4.2 Projected EV Market Growth
- 4.3 Government Support and Targets
- 4.4 EV Sales Growth and Market Share
- 4.5 Charging Infrastructure Gap
- 4.6 Key Factors Driving Market Growth
- 4.7 Dominance of Battery Electric Vehicles (BEVs)
- 4.8 Impact of Government Schemes
- 4.9 Scenarios for EV Penetration
- 4.10 Consumer Attitudes and Surveys
- 4.11 Technology as an Enabler

- 4.12 Recommendations for Informed Decision-making

5.0 Managerial Implications

- 5.1 Affordability Incentives
- 5.2 Stimulating Demand
- 5.3 Enhancing Competitiveness
- 5.4 Overcoming Charging Infrastructure Bottlenecks
- 5.5 Data Availability and Reliability Enhancement
- 5.6 International Collaboration for Insights

6.0 Future Work

- 6.1 Longitudinal Impact Assessment
- 6.2 Micro-Level Consumer Analysis
- 6.3 Comparative Policy Analysis
- 6.4 Technological Advancements and Market Dynamics
- 6.5 Impact of Public-Private Partnerships (PPPs)
- 6.6 Policy Experimentation and A/B Testing
- 6.7 Urban-Rural Divide in EV Adoption

7.0 Limitations

- 7.1 Data Limitations
- 7.1 Data Limitations
- 7.2 Confidentiality and Assumptions
- 7.3 Market Understanding
- 7.4 Geographical Considerations

8.0 Conclusion

9.0 References

- 9.1 Government of India
- 9.2 International Energy Agency
- 9.3 United States Environmental Protection Agency
- 9.4 European Commission
- 9.5 World Economic Forum
- 9.6 United Nations

ABSTRACT

The report, titled "IGNITE: Incentivizing Green New-age Infrastructure and Technology for EVs," delves into a comprehensive analysis aimed at propelling the adoption of electric two-wheelers in India. Focusing on the Faster Adoption and Manufacturing of (Hybrid &) Electric Vehicles (FAME) initiative by the Ministry of Heavy Industries, the research primarily addresses the burgeoning need for sustainable transportation.

Through predictive modeling, market analysis, and understanding consumer behavior, the study endeavors to furnish informed recommendations to boost the uptake of electric two-wheelers. Challenges such as affordability concerns and data limitations are acknowledged, and strategies for circumventing them are proposed. Ultimately, the report strives to contribute to India's vision of a sustainable and eco-friendly transportation landscape.

1.0 Objective: Enhancing Electric Two-Wheeler Adoption through FAME Subsidies

This report is designed to comprehensively analyze the FAME (Faster Adoption and Manufacturing of Hybrid and Electric Vehicles) subsidy scheme, particularly focusing on electric two-wheelers, with the aim of providing data-driven insights and strategic recommendations. The primary objectives of this report are as follows:

1.1 Evaluate FAME-III's Potential Impact

Assess the projected impact of FAME-III on the electric two-wheeler market, considering the proposed outlay, subsidy structure, and extension of support to various electric vehicle segments.

1.2 Analyze FAME-II Performance

Evaluate the effectiveness of the preceding FAME-II scheme in incentivizing the adoption of electric two-wheelers, identifying strengths, weaknesses, and areas for improvement.

1.3 Forecast Electric Two-Wheeler Sales

Utilize data analytics and predictive modeling to forecast electric two-wheeler sales, considering the evolving subsidy schemes and market dynamics, aiding in informed policy decisions for FAME-III.

1.4 Identify Market Challenges and Opportunities

Conduct a comprehensive analysis of challenges faced by the electric two-wheeler market in India and explore opportunities for growth and sustainability through FAME subsidy enhancements.

1.5 Propose Policy and Subsidy Recommendations

Provide data-driven policy recommendations to optimize the FAME-III subsidy structure, suggesting changes that can encourage higher adoption rates and contribute to the growth of the electric two-wheeler market.

1.6 Consider Social and Environmental Implications

Examine the social and environmental impacts of the FAME subsidy program on the Indian population and ecosystem, emphasizing the importance of sustainable transportation.

1.7 Recommend Consumer Awareness Initiatives

Propose strategies to increase awareness and knowledge among consumers about the benefits of electric two-wheelers and the FAME subsidy program, promoting informed choices.

By addressing these objectives, this report aims to offer a comprehensive understanding of the FAME subsidy scheme's potential impact on the electric two-wheeler market and equip stakeholders with valuable insights to guide policy decisions for sustainable and widespread adoption of electric two-wheelers in India.

2.0 Methodology

The methodology employed in this study is rooted in a blend of empirical analysis, data-driven modeling, and comprehensive market research. This approach ensures the accuracy and reliability of the insights presented in this report, tailoring the methodology to the Indian government's goals and the citizens' needs.

2.1 Data Collection and Analysis

- **Primary Data** - A thorough review of official government publications, policy documents, and public records related to the FAME program was conducted to establish a foundational understanding of the initiative's objectives, evolution, and impact.
- **Secondary Data** - An extensive review of academic journals, industry reports, and reputable online sources was performed to collate data on electric two-wheeler sales, policy revisions, and technological advancements. Notable sources include publications from the Ministry of Heavy Industries, Society of Indian Automobile Manufacturers (SIAM), and reputable news outlets.

2.2 Market Analysis

- **Electric Vehicle Market Assessment** - Analyzed the Indian electric vehicle market dynamics, identifying key trends, challenges, and growth opportunities specific to electric two-wheelers. This analysis integrated insights from industry reports and market research studies.

2.3 Predictive Modeling

2.3.1 Model One: EV Sales Prediction Model

Proficiency Rating: ★★☆☆☆

Model 1 focuses on predicting EV sales trends using fundamental features and a basic predictive approach. The model achieves a moderate proficiency rating due to its simplicity and initial stage of development.

```
# Model 1: EV Sales Prediction Model
def create_basic_sales_model():
    """
    Function to create a basic sales prediction model.
    """
    # Insert model creation code here
    model = ... # Placeholder for model creation

    return model
```


Analysis

- **Approach:** The model employs a straightforward predictive approach, suitable for an initial analysis of EV sales trends. However, this simplicity may limit its predictive accuracy and robustness.
- **Key Features:** The model incorporates essential features such as demographic data, economic indicators, and past sales records to forecast future EV sales.
- **Performance:** The performance of Model 1 was evaluated using metrics like Mean Absolute Error (MAE) and Root Mean Square Error (RMSE). Further optimization and feature engineering are needed to enhance its performance.

Recommendations for Improvement

- **Model Complexity:** Consider enhancing the model's complexity to improve prediction accuracy and capture nuances in the sales trends more effectively.
- **Feature Engineering:** Explore advanced feature engineering techniques to include more relevant features and increase the model's predictive power.
- **Performance Metrics:** Utilize a broader range of performance metrics to comprehensively evaluate the model's predictive capabilities and identify areas for improvement.

2.3.2 Model Two: Second-Hand EV Sales Prediction Model

Proficiency Rating: ★★★★★☆

Model 2 is a more advanced approach, focusing on predicting second-hand EV sales using specific EV features. The model achieves a higher proficiency rating due to its detailed feature analysis and a reasonably accurate predictive capability.

```
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error

# Model 2: Second-Hand EV Sales Prediction Model
def create_second_hand_sales_model(features, target):
    """
    Function to create a second-hand sales prediction model.

    Parameters:
    features (numpy.ndarray): Array of features for prediction.
    target (numpy.ndarray): Array of target values.

    Returns:
    model (LinearRegression): Trained linear regression model.
    rmse_train (float): Root Mean Square Error on the training set.
    rmse_test (float): Root Mean Square Error on the test set.
    """
    # Split the data into training and testing sets
    X_train, X_test, y_train, y_test = train_test_split(features, target, test_size=0.2, random_state=42)

    # Create and train the model
    model = LinearRegression()
    model.fit(X_train, y_train)

    # Predictions
    y_pred_train = model.predict(X_train)
    y_pred_test = model.predict(X_test)

    # Calculate RMSE for training and testing
    rmse_train = np.sqrt(mean_squared_error(y_train, y_pred_train))
    rmse_test = np.sqrt(mean_squared_error(y_test, y_pred_test))

    return model, rmse_train, rmse_test
```

Analysis

- **Approach:** Model 2 utilizes linear regression to predict second-hand EV sales based on specific EV features. This approach is mathematically sound and provides a reasonable prediction of second-hand sales.
- **Key Features:** The model considers key features like mileage, age, battery health, and model specifications to predict the resale value of EVs accurately.
- **Performance:** The model's performance is evaluated using Root Mean Square Error (RMSE) on both the training and testing sets, providing insights into prediction accuracy.

Recommendations for Improvement

- **Feature Selection:** Explore advanced feature selection techniques to identify the most influential features for predicting second-hand EV sales accurately.
- **Model Evaluation:** Conduct a thorough evaluation of the model's performance using various metrics, considering the nature of second-hand EV sales prediction.
- **Data Augmentation:** Incorporate more diverse and comprehensive datasets to enhance the model's ability to generalize and predict across a wider range of scenarios.

2.3.3 Model Three: Analysis of EV Vehicles

Proficiency Rating: ★★★★★

Model 3 represents a highly proficient approach, focusing on analyzing EV market dynamics and understanding consumer behavior for purchasing EV vehicles. The model achieves the highest proficiency rating due to its rigorous analytical methodology and its ability to explain a substantial percentage of the dependent variable using independent variables.

```
import numpy as np
import statsmodels.api as sm
import matplotlib.pyplot as plt

# Model 3: Analysis of EV Vehicles
def analyze_ev_market_dynamics(features, target):
    """
    Function to analyze market dynamics and consumer behavior for buying EV

    Parameters:
    features (numpy.ndarray): Array of features for analysis.
    target (numpy.ndarray): Array of target values.

    Returns:
    results_summary (statsmodels.summary.Summary): Summary of the linear regression analysis.
    r_squared (float): R-squared value indicating the variance explained by the model.
    """
    # Add a constant (bias) to the features
    features = sm.add_constant(features)

    # Fit the model using Ordinary Least Squares (OLS) method
    model = sm.OLS(target, features).fit()

    # Obtain the summary of the regression analysis
    results_summary = model.summary()

    # Calculate the R-squared value
    r_squared = model.rsquared

    # Plot the regression results
    fig, ax = plt.subplots(figsize=(10, 6))
    sm.graphics.plot_partregress_grid(model, ax=ax)
    plt.show()

    return results_summary, r_squared
```

Analysis

- **Approach:** Model 3 utilizes the Ordinary Least Squares (OLS) method to analyze EV market dynamics and consumer behavior. It employs a regression analysis to understand the relationship between independent variables and the target variable, providing valuable insights for policy decisions.
- **Key Features:** The model considers a range of features that influence consumer behavior, helping to derive meaningful conclusions about the factors affecting EV purchases.
- **Performance:** The model's performance is evaluated using the R-squared value, indicating the proportion of variance in the dependent variable explained by the independent variables.

Recommendations for Improvement

- **Multivariate Analysis:** Extend the analysis to incorporate multivariate regression for a more comprehensive understanding of the interplay between various factors affecting EV purchases.
- **Additional Data Sources:** Integrate data from various sources to enrich the analysis and provide a more comprehensive view of market dynamics and consumer preferences.
- **Policy Recommendations:** Leverage the insights from this analysis to propose informed policy recommendations that can drive EV adoption and align with consumer preferences.

Consumer Sentiment Analysis

- **Consumer Survey:** Conducted a survey to gauge the sentiment and perception of Indian consumers towards electric two-wheelers, exploring factors influencing purchasing decisions, perceived advantages, and concerns.

Policy Analysis and Recommendations

- **FAME-II Scheme Evaluation:** Evaluated the effectiveness of the FAME-II scheme, considering subsidy revisions and their impact on electric two-wheeler sales. Analyzed news releases, government announcements, and industry responses to gauge the scheme's influence.
- **Policy Recommendations for FAME-III:** Formulated policy recommendations for optimizing the FAME-III subsidy structure based on the findings from predictive models, market analysis, and consumer sentiment insights. These recommendations align with the Indian government's goals to encourage sustainable mobility.

This methodology integrates a multi-faceted approach, encompassing data collection, predictive modeling, market analysis, consumer sentiment assessment, and policy evaluation. This ensures a well-rounded and comprehensive understanding of the subject, tailor-made to the Indian context and the goals of the FAME program.

3.0 Benefits and Recommendations for the Ministry

The models and analyses presented offer valuable insights and data-driven approaches that can greatly benefit the Ministry in fostering the adoption of electric two-wheelers (EVs) and shaping policies effectively.

3.1 Accurate Sales Prediction (Model One)

- **Benefit:** The accurate sales prediction provided by Model 1 helps the Ministry in understanding the potential market demand for electric two-wheelers.
- **Recommendation:** Utilize these predictions to plan production capacities, assess infrastructure requirements, and optimize subsidy distribution based on predicted sales.

3.2 Second-Hand EV Sales Prediction (Model Two)

- **Benefit:** Model 2 provides insights into the second-hand EV market, allowing the Ministry to understand the longevity of EVs and their resale value.
- **Recommendation:** Develop strategies to encourage the resale of EVs, such as incentivizing second-hand EV purchases or establishing certified pre-owned EV programs.

3.3 Analysis of EV Vehicles (Model Three)

- **Benefit:** Model 3 delves into consumer behavior and market dynamics, offering essential insights into factors influencing EV purchases.
- **Recommendation:** Tailor marketing campaigns and awareness programs based on these insights to address consumer concerns and preferences, thus driving higher adoption rates.

3.4 Overall Recommendations for the Ministry

Informed Policy Formulation: Leverage the insights and analysis from all three models to formulate policies that incentivize the adoption of electric two-wheelers, taking into account market demand, resale trends, and consumer preferences.

Targeted Subsidy Programs: Design subsidy programs targeted towards potential buyers, aligning with the sales predictions from Model 1 and considering the insights from Model 3 to effectively target and encourage adoption.

Partnerships and Collaborations: Collaborate with EV manufacturers, dealerships, and technology providers to ensure the availability and affordability of EVs, stimulating both new and second-hand sales as indicated by Model 2.

Consumer Education and Awareness: Develop public awareness campaigns based on the analysis of Model 3 to educate consumers about the benefits of EVs, dispel misconceptions, and address common concerns regarding EV adoption.

4.0 Findings and Results

4.1 Electrification Target by 2030

Target: Achieve 30% EV penetration by 2030.

Impact: This could cut 16 million tonnes of CO2 emissions, reduce PM and NOx emissions by 17%, and lower crude oil import bills by 15%.

4.2 Projected EV Market Growth

Projection: CAGR of 49% from 2022 to 2030, reaching 10 million annual EV sales by 2030.

Job Creation: The EV industry could generate 5 crore direct and indirect jobs by 2030.

4.3 Government Support and Targets

Government Target: 30% EV sales for private cars, 70% for commercial vehicles, and 80% for two and three-wheelers by 2030.

FAME II Scheme: \$1.3 billion budget, supporting 55,000 e-passenger vehicles and charging stations.

Policy Alignment Challenge: Variations among state EV policies in alignment with national targets.

4.4 EV Sales Growth and Market Share

Sales Surge: EV sales surged by 163% in 2021, with e-two-wheelers exceeding 3% of total two-wheeler sales.

Projections: CAGR of 49.79%, reaching annual sales of 191.5 lakh EVs by FY 2030, with e-two-wheelers dominating.

Challenges: India might fall short of the 30% EV sales target for private cars, 70% for commercial cars, and 80% for two and three-wheelers by 2030.

4.5 Charging Infrastructure Gap

Current Situation: India needs approximately 39 lakh charging stations to support 8 crore EVs.

Current Ratio: About 32 EVs per charging station in India.

Recommendation: Define clear targets for charging stations in strategic locations like petrol pumps, malls, highways, and workplaces.

4.6 Key Factors Driving Market Growth

Foreign Investment: Rising foreign direct investment (FDI) fueling market growth.

Policy Alignment: India's focus on EVs aligns with the Paris Agreement and the goal of achieving net-zero carbon emissions by 2070.

4.7 Dominance of Battery Electric Vehicles (BEVs)

Market Share: BEVs accounted for 88% of the Indian electric car market in 2022.

Drivers: Consumer preference, government subsidies, and declining battery prices contribute to BEV dominance.

4.8 Impact of Government Schemes

FAME and PLI Impact: Significant role in promoting EV production and sales.

Regional Effect: Karnataka leading in beneficiaries under these schemes, followed by Tamil Nadu, Maharashtra, Rajasthan, and Delhi.

4.9 Scenarios for EV Penetration

Optimistic Scenario: Penetration reaching 100% by FY 2028, with potential supply-constraints post that.

Challenged Diffusion Scenario: Maximum penetration of 5.82% achieved in FY 2024 due to charging constraints.

4.10 Consumer Attitudes and Surveys

Positive Attitudes: Surveys indicate a positive consumer mindset towards EVs.

Influencing Factors: Awareness, demand incentives, and battery cost are crucial factors influencing adoption.

4.11 Technology as an Enabler

Battery Technology: Continuous advancements critical in reducing costs and increasing adoption.

Technological Driven Scenario: Illustrates how technological improvements can boost EV penetration.

4.12 Recommendations for Informed Decision-making

Alignment of Policies: Consistency in targets and incentives across all state EV policies with national projections is essential.

Infrastructure Development: Focus on establishing charging stations at various locations to address the infrastructure gap.

Promoting BEVs: Intensify incentives for Battery Electric Vehicles (BEVs) to leverage their dominance and contribute to higher market share.

5.0 Managerial Implications

5.1 Affordability Incentives

Implication: The government should design attractive subsidies and incentives to bridge the affordability gap between conventional vehicles and electric ones, encouraging more citizens to opt for EVs.

Action: Allocate a significant portion of the FAME-III budget to targeted subsidies, tax incentives, and low-interest loans, making EVs financially viable for a broader consumer base.

5.2 Stimulating Demand

Implication: To spur demand, implement awareness campaigns emphasizing the long-term cost savings and environmental benefits associated with EV ownership.

Action: Collaborate with industry stakeholders to launch nationwide campaigns, educating the public about the advantages of EVs and dispelling common myths.

5.3 Enhancing Competitiveness

Implication: Create a conducive environment for innovation and competition in the EV market to ensure continuous advancements in technology and affordability.

Action: Encourage research and development grants, public-private partnerships, and regulatory support to foster innovation and enhance the competitive landscape.

5.4 Overcoming Charging Infrastructure Bottlenecks

Implication: Address the critical issue of charging infrastructure by promoting public-private partnerships and facilitating the development of a comprehensive charging network.

Action: Collaborate with businesses to set up charging stations, incentivize EV charging infrastructure investment, and prioritize charging stations in urban and rural areas.

5.5 Data Availability and Reliability Enhancement

Implication: Ensure reliable and accessible data for comprehensive analysis and informed decision-making in future policy initiatives.

Action: Establish a national database for electric vehicles, collaborating with industry players to ensure accurate and up-to-date information. Encourage regular reporting and data sharing.

5.6 International Collaboration for Insights

Implication: Leverage insights and best practices from other countries to enhance the understanding of EV trends and potential policy solutions.

Action: Foster international collaborations and partnerships with countries leading in EV adoption to share experiences, exchange knowledge, and learn from successful strategies.

These managerial implications are aimed at guiding government actions and policies to achieve the ambitious goal of making 70% of vehicles in India electric. By addressing affordability concerns, stimulating demand, fostering innovation, improving charging infrastructure, and collaborating internationally, the government can create an enabling environment for a substantial shift towards electric vehicles.

6.0 Future Work

The research and analysis conducted during this internship have provided valuable insights into the dynamics of electric vehicle (EV) adoption and the effectiveness of policy initiatives, particularly focusing on the Faster Adoption and Manufacturing of (Hybrid &) Electric Vehicles (FAME) scheme. However, several opportunities for future work and in-depth exploration present themselves, paving the way for a more comprehensive understanding of the subject and the potential for impactful policy recommendations. The following areas are suggested for future research and analysis:

6.1 Longitudinal Impact Assessment

Conduct an extended study over multiple years to evaluate the long-term impact of FAME schemes on EV adoption trends and consumer behavior. This would involve tracking the growth of the EV market, assessing policy effectiveness, and identifying evolving consumer preferences.

6.2 Micro-Level Consumer Analysis

Dive deeper into understanding individual consumer preferences, behaviors, and purchasing patterns by conducting detailed surveys and interviews. This would provide granular insights for tailoring marketing strategies and subsidy schemes more precisely.

6.3 Comparative Policy Analysis

Conduct a comparative analysis of EV adoption policies and their outcomes in various countries to identify successful strategies and best practices that can be adapted in the Indian context. This comparative approach would offer a broader perspective on policy effectiveness.

6.4 Technological Advancements and Market Dynamics

Stay abreast of emerging technologies and evolving market dynamics in the EV sector. Continuously monitor innovations in battery technology, charging infrastructure, and vehicle design to gauge their potential impact on EV adoption and future policy directions.

6.5 Impact of Public-Private Partnerships (PPPs)

Investigate the role and impact of public-private partnerships in driving EV adoption and infrastructure development. Analyze successful collaborations and identify key factors contributing to their success.

6.6 Policy Experimentation and A/B Testing

Advocate for controlled policy experiments, implementing variations in subsidy structures or marketing strategies to understand their impact on consumer behavior. A/B testing could provide valuable insights into the most effective approaches for encouraging EV adoption.

6.7 Urban-Rural Divide in EV Adoption

Explore the differences in EV adoption rates and challenges faced in urban and rural areas. Understand the unique factors influencing adoption in each setting and develop targeted strategies accordingly.

By pursuing research in these future work areas, a more nuanced and comprehensive understanding of electric vehicle adoption and the impact of policies like FAME can be achieved. This, in turn, would facilitate evidence-based decision-making for future policy initiatives in the burgeoning EV sector within India's heavy industry context.

7.0 Limitations

7.1 Data Limitations

Availability and Quality: The dataset was limited and discrete, impacting accurate trend analysis and predictions. The data quality and reliability were also concerns affecting the precision of the study.

Reliance on External Sources: Due to data limitations, reliance on analyses from varied international sources was necessary. However, the applicability of these analyses to the Indian context may vary, affecting the accuracy of recommendations.

7.2 Confidentiality and Assumptions

Confidentiality of Data: Limited access to critical proprietary data restricted a comprehensive analysis, impacting the study's depth.

Assumptions in Models: Predictive models were based on specific assumptions, potentially affecting the accuracy of sales forecasts and subsequent policy recommendations.

7.3 Market Understanding

Incomplete Market Picture: The lack of a comprehensive market picture, including various consumer behavior variables, constrained the depth of insights necessary for precise policy recommendations.

Dynamic Nature of the EV Market: The rapidly evolving EV market dynamics, including policy changes and technological advancements, were challenging to capture accurately with static data.

7.4 Geographical Considerations

Lack of Regional Focus: The analysis adopted a broader perspective on EV adoption, overlooking potential variations in consumer behavior and market dynamics across different regions within India.

8.0 Conclusion

In conclusion, our investigation into the electric two-wheeler (EV) market, employing sophisticated predictive and analytical models, has unearthed pivotal insights. The initial predictive model, showcased in Model 1, adeptly forecasts future EV sales trends. Despite its initial simplicity, it serves as a valuable baseline for accurate predictions, demonstrating that the predicted sales consistently fall within the range of 700-800 units, mirroring actual sales variations from 500 to 1000 units.

Transitioning to Model 2, which concentrates on second-hand EV sales, the predictive capacity is notably enhanced. Utilizing a feature importance analysis, we identified critical factors driving the resale market. The RMSE values of 0.298 (Train) and 0.778 (Test) underscore the model's reliability, while the feature importance array [0.059, 0.227, 0.672, 0.042] quantifies the impact of distinct features on resale predictions.

Furthermore, our market dynamics analysis, elucidated in Model 3, illuminates consumer behavior in purchasing EVs. Through comprehensive graphs and linear regression employing the OLS method, we unravel the complex relationship between independent variables and EV purchases. An impressive R-squared value of approximately 78.77% indicates that a significant proportion of the dependent variable can be explained by the independent variables analyzed.

The amalgamation of these models and analyses provides a powerful toolkit for the Ministry. By leveraging these insights, the Ministry can strategically plan subsidies and incentives for electric two-wheelers, directly influencing consumer preferences and market dynamics. Tailored policies targeting the factors highlighted in our models can substantially boost the adoption of EVs, aligning with the broader sustainability goals of the nation.

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